Supplementary Material for the Article: Playing Technique Recognition via Time–Frequency Scattering

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This document provides further details regarding datasets and detection results for our work "Playing Technique Recognition via Time–Frequency Scattering" [1].

A. Existing Computational Research on Playing Techniques

Besides the summary of existing computational research on playing techniques in the our paper [1], we provide a complete list of playing techniques analysed and the corresponding methodologies applied in Table I.

B. Additional Information on the Datasets

Table II displays number of ground truth frames, total duration of the ground truth, and duration of co-articulations for each playing technique in the CBFdataset. As can be seen, the number of samples for each type of technique is highly imbalanced. Portamento has 1010 samples while the number of trill is only 197. As compared to the duration of the whole dataset of 157 minutes, that of the co-articulations only amount to 5 minutes. To have comparable result to the extended datasets discussed in our paper [1] and to obtain a confusion matrix between different CBF techniques, we discard the co-articulations samples and conduct a multiclass classification.

Fig. 1 shows the comparison between glissando played in isolation and that performed in full-length pieces. The isolated one consists of consecutive note changes with approximately equal duration, while the performed one vary from each other in different ways. Player 1 lengthened the first note for the glissando at position 1, while emphasised the beginning of the glissando at position 2 by increasing the intensity. Player 2 performed the glissando together with flutter-tongue. These variabilities necessitate the data collection of playing techniques in music context.

To have a better understanding of the detection result in the datasets with playing techniques accurately annotated in real-world pieces, we plot the boxplots of the duration of each type of technique in the CBFdataset and VPset. As can be seen in Fig. 2, pitch evolution-based techniques (PETs) exhibit shorter duration than that of pitch modulation-based techniques (PMTs).

C. Additional Result Analysis

Fig. 3 and Fig. 4 are the confusion matrices obtained for the SOL dataset and VocalSet. The recognition for the SOL dataset

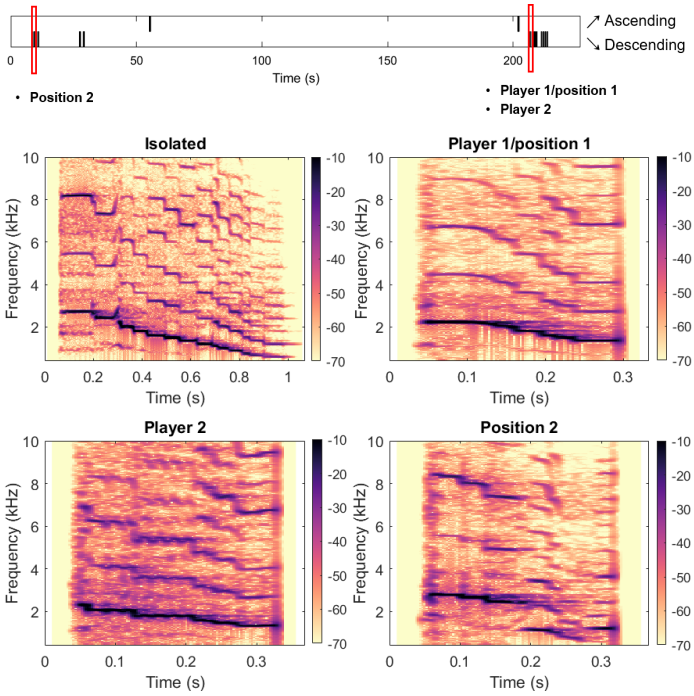

Fig. 1. Variations of glissandi between different players and different temporal positions in the performance of *Busy Delivery Harvest*.

presents few confusions between techniques. This may be attributed to the dataset structured with short clips of individual techniques. No technique is based on the same note of the same instrument, which offers high intra-class variability for playing techniques. As can bee seen from the normalised confusion matrix in Fig. 4 (b), the lip trill obtains the best score while the inhaled technique obtains the lowest. There exists a large amount of confusion between breathy and vocal fry. The straight technique exhibits high confusion with other techniques.

References

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TABLE I
SUMMARY OF EXISTING COMPUTATIONAL RESEARCH ON PLAYING TECHNIQUES

Instrument	Expressive patterns	Methodologies applied	Cite
Guitar	Ornamentation defined by rules	Dynamic time warping sequence matching	[2]
	Legato, glissando	Symbolic, aggregate approximation	[3]
	Muted, harmonic, bend, slide, hammer-on, pull-off	Rule-based detection	[4]
	Muting, vibrato, pull-off, hammer on, sliding, bending	Sparse cepstral and phase codes	[5]–[7]
	Plucking: finger, picked, muted, slap; expression: harmonics, vibrato, bending, slide, etc.	Feature extraction, support vector machines with radial basis function kernel	[8], [9]
Piano	Trills	Independent component analysis	[10]
	Pdealling techniques	Supporter vector machines	[11]
Violin	Beginners' faults on playing techniques	K-nearest neighbours	[12]
	Flageolet, non-vibrato, pizzicato, sordino, spiccato, sul ponticello, sul tasto, tremolo	Sparse modelling	[13]
Drums	strike, buzz roll, drag, flam	Non-negative matrix factorization-based activation	[14]
	A variety of drum playing techniques	K-nearest neighbours, partial decision tree, etc.	[15]
Cello	Vibrato, tremolo, trill, glissando, etc.	Folded convolutional neural network	[16]
Irish flute	Cut, strike, crann, roll, shake	Rule-based method; hidden Markov models (HMMs)	[17]
Highland pipe	60 ornamentations	Dynamic time warping	[18]
Ney	Vibrato, kaydirma	Rule-based characterisation	[19]
Erhu	Vibrato, portamento	Filter diagonalisation method, HMMs	[20]
Guqin	None, vibrato, upward/downward/inverted-U/U protamento	Neural network	[21]
Chinese bamboo flute	Vibrato, tremolo, till, flutter-tongue, acciacatura, portamento, glissando	HMMs, adaptive/joint time–frequency scattering	[22]–[24]
	Trill, flutter-tongue, tremolo	Synthesising	[25], [26]
Other	Vibrato, glissando	Rule-based methods	[27]
Multi-instruments	140 instrumental playing techniques (isolated)	Scattering transform	[28]
Vocal techniques	Vibrato, glissando, linear note change	COSFIRE filter	[29]
	17 different vocal techniques	VocalSet	[30]

TABLE II
NUMBER OF SAMPLES, DURATION OF GROUND TRUTH, AND DURATION OF CO-ARTICULATIONS FOR THE SEVEN TECHNIQUES IN THE CBFDATASET. TOTAL DURATION OF THE CBFDATASET IS 157 MINUTES, INCLUDING 5-MINUTE CO-ARTICULATIONS

Techniques	Flutter-tongue	Trill	Vibrato	Tremolo	Acciacatura	Portamento	Glissando
Number of samples	628	197	519	555	441	1010	490
Duration of ground truth (minute)	7.9	15.0	13.8	6.0	1.7	6.3	5.0
Duration of co-articulations (minute)	1.3	0.1	3.7	3.7	0	0.9	0.3

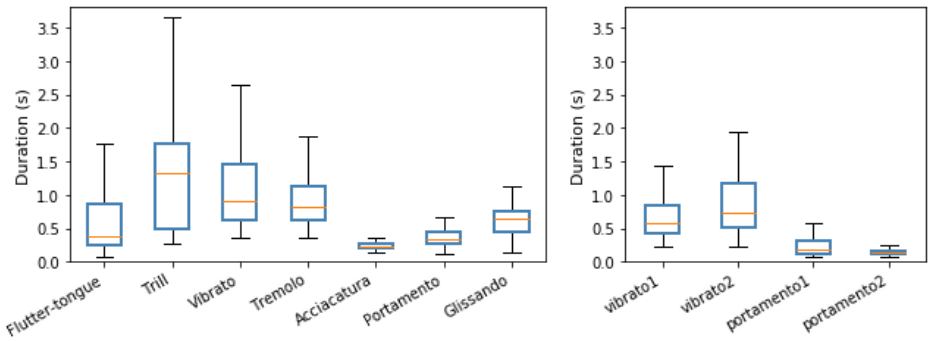
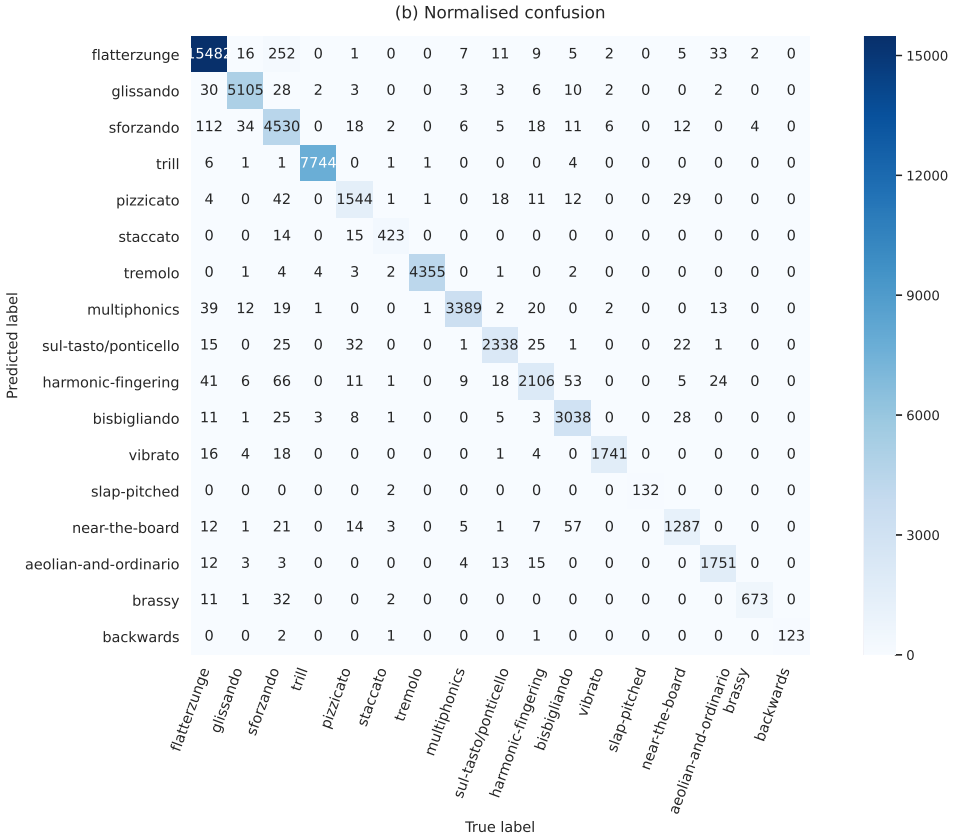


Fig. 2. Boxplots of duration for playing techniques in the CBFdataset and VPset. Top: seven playing techniques in the CBFdataset; bottom: vibrato and portamento in the VPset (vibrato1=vibrato of erhu and violin; vibrato2=vibrato of instrumental excerpts; portamento1 = portamento of erhu and violin; portamento1 = portamento of Beijing opera singing).



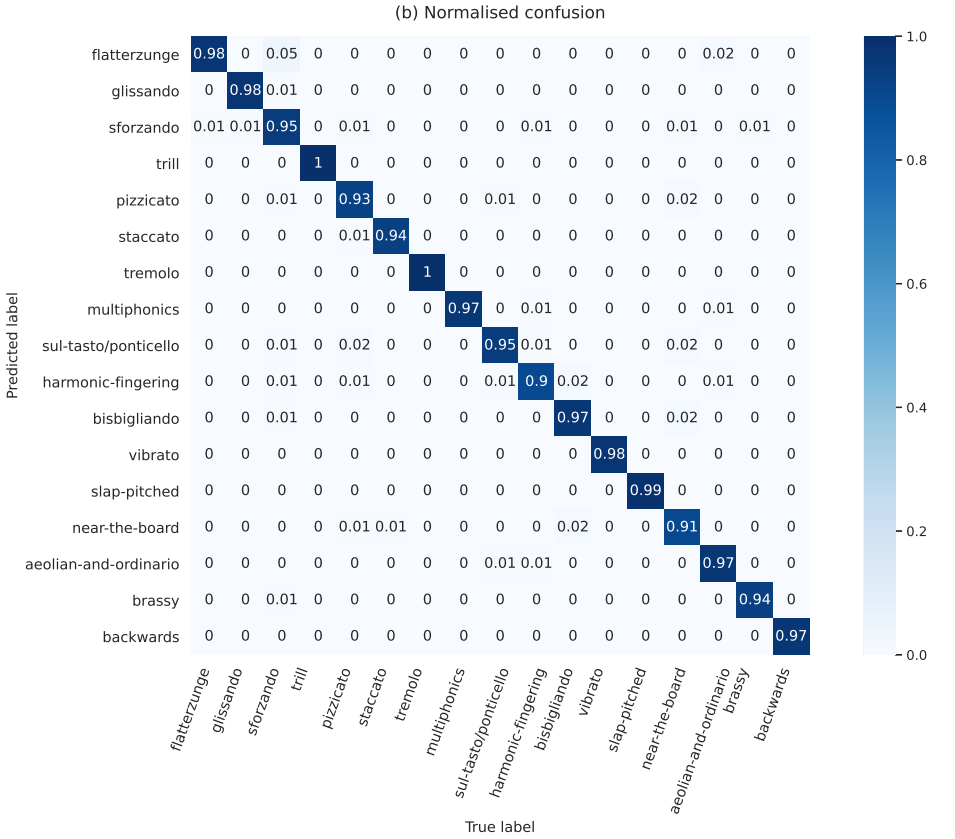


Fig. 3. Confusion matrices obtained for multi-class classification of techniques in the SOL dataset. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

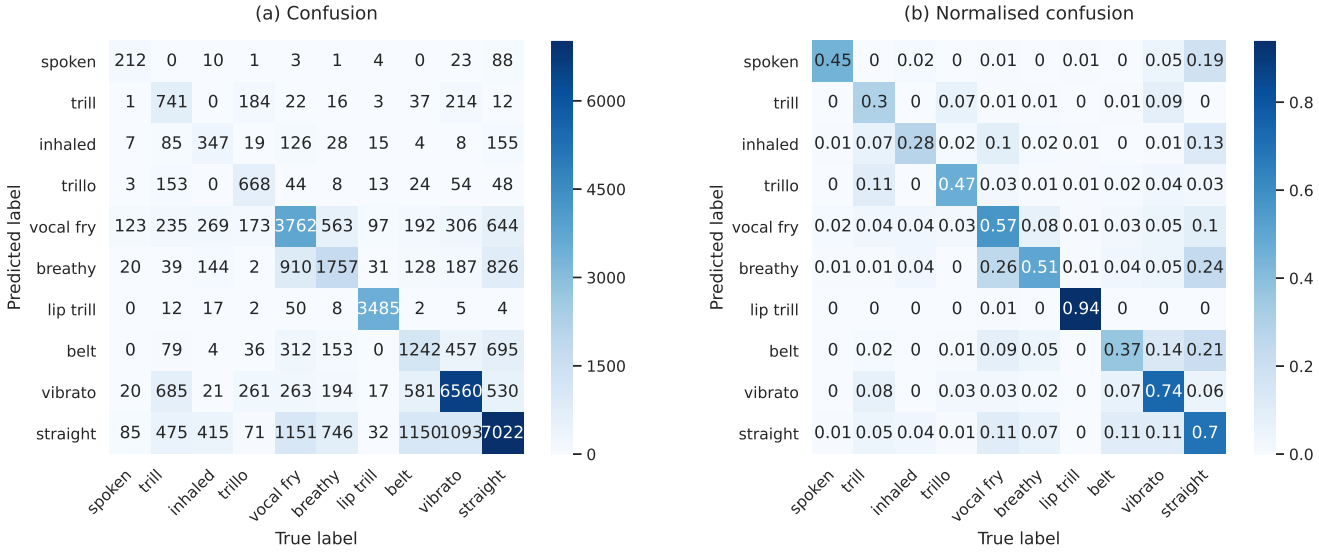


Fig. 4. Confusion matrices obtained for multi-class classification of ten vocal techniques. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

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